

ROBIN ROY

Full-Stack Software Engineer — Python · Django · FastAPI · React.js · PostgreSQL | Generative AI

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SUMMARY

Full-stack Software Engineer with experience building and operating statewide government applications (Government of Kerala, India) used daily by 1,500+ officers. Experienced in shipping secure REST APIs in Python (Django, FastAPI), responsive React.js frontends, and Generative AI features using LLMs, Retrieval-Augmented Generation (RAG), and vector search. Hands-on with PostgreSQL, Docker, AWS EC2, NGINX, and Agile delivery; holds an M.Tech (Robotics & Automation) with an IEEE publication.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, SQL

Generative AI & LLMs: Agentic AI, LangChain, LLM API Integration (Google Gemini, Groq), Retrieval-Augmented Generation (RAG), Semantic Search, Prompt Engineering, Vector Embeddings (Sentence Transformers), ChromaDB

Backend: FastAPI, Django, Django REST Framework (DRF), REST APIs, SQLAlchemy, Pydantic, JWT Authentication, OAuth2, Microservices, Spring Boot (working knowledge)

Frontend: React.js, React Hooks, Redux, Material UI (MUI), Tailwind CSS, Bootstrap, HTML5, CSS3

Databases: PostgreSQL, MongoDB, MySQL, SQLite, ChromaDB (Vector Database)

Cloud & DevOps: Docker, Docker Compose, AWS EC2, NGINX, CI/CD, Git, GitHub, Postman, Swagger/OpenAPI

Software Engineering & Testing: System Design, API Design, Unit Testing, pytest, Code Reviews, Agile (Scrum)

WORK EXPERIENCE

Jr. Software Engineer

Sep 2024 - Present

CDTI, Digital University Kerala (DUK), Thiruvananthapuram, Kerala, India

- Built and maintained 2 Government of Kerala applications end to end — React.js frontends and Python/FastAPI + PostgreSQL backends — used daily by 1,500+ officers and staff of the Department of Economics and Statistics across all 14 districts
- Developed secure REST APIs with JWT authentication and role-based access control across 5+ user roles, enabling safe frontend–backend communication for sensitive government data
- Conducted applied Generative AI research, building Mirai — a RAG-based document Q&A backend (FastAPI, Google Gemini, ChromaDB) — to evaluate document intelligence for departmental workflows
- Built responsive, role-aware frontend interfaces in React.js (Hooks, Redux, Material UI, Tailwind CSS), developing reusable components shared across both applications
- Designed and optimized PostgreSQL schemas for statewide statistical and agricultural datasets, improving key query performance by 35% through indexing and relational restructuring
- Containerized services with Docker/Docker Compose and deployed to AWS EC2 behind NGINX; documented all APIs with Swagger/OpenAPI for team-wide consumption
- Contributed to the team's move toward microservices: API integration, debugging, code reviews, and unit testing in Agile sprints; also delivered features in Django and Spring Boot where projects required

Full-Stack Development Trainee

Mar 2023 – Aug 2024

Brototype, Kochi, Kerala, India

- Transitioned into software engineering through a rigorous, self-driven program built on daily coding tasks, weekly project reviews, peer learning, and mock technical interviews
- Built and deployed 10+ portfolio projects, including Zenix (Django e-commerce with Razorpay payments, deployed on AWS EC2) and ConnectVerse (Django REST Framework + React social platform) — github.com/robinsamuelroy

FEA Software Engineer Intern

Mar 2021 - Feb 2022

Dimension Design Analysis, Amrita TBI, Kollam, Kerala, India

- Automated preprocessing, data validation, and analysis workflows using Python, integrating CAD models with ANSYS simulation pipelines

- Built reusable simulation-data processing tools adopted across the engineering team, reducing manual analysis steps and improving workflow consistency

Quality Assurance Engineer

Nov 2018 - Mar 2020

Orifice MEP LLP, Kochi, Kerala, India

- Coordinated quality control across MEP projects, tracking isometric drawings and implementing corrective-action protocols aligned with industry standards

PROJECTS

AIDEA – EARAS — Agricultural Statistics Platform, Govt. of Kerala

- Developed core services for the EARAS module of AIDEA, the statewide platform replacing the department’s legacy DESCAS system for agricultural statistics
- Built the user service for authentication and authorization — hierarchical role-based access from field investigators to district and headquarters officers, employee lifecycle management, and OTP-based credential flows
- Developed the BTR service integrating external land-records data through a shared government API, including annual data generation with approval workflows
- Built the Tour Diary service end to end, enabling investigators to submit tour diaries and progress reports through multi-level review and approval

Technologies: Spring Boot, React.js, Django, PostgreSQL, Microservices, AWS EC2

Mirai — AI Knowledge Assistant | github.com/robinsamuelroy/mirai-ai-backend

- Built an end-to-end RAG pipeline with FastAPI: PDF upload and text extraction, automatic chunking, local embedding generation (Sentence Transformers), semantic search over ChromaDB, and Gemini-grounded answer generation
- Implemented JWT authentication with multi-user data isolation, conversation memory, and persistent chat history in PostgreSQL
- Designed a layered Router → Service → Repository architecture for testability; containerized with Docker and deployed on AWS EC2 behind NGINX, with a React.js chat interface in development

Technologies: Python, FastAPI, React.js, Google Gemini API, RAG, ChromaDB, Sentence Transformers, PostgreSQL, SQLAlchemy, JWT, Docker, NGINX, AWS EC2

Zenix — E-Commerce Platform | github.com/robinsamuelroy/zenix-ecommerce

- Developed a full e-commerce platform handling 4 core modules (inventory, orders, coupons, payments) with an optimized Django ORM architecture
- Integrated Razorpay payment gateway and deployed to production on AWS EC2 with NGINX

Technologies: Django, PostgreSQL, Razorpay, JavaScript, Bootstrap, AWS EC2

EDUCATION

Master of Technology, Robotics and Automation

2020 - 2022

Amrita University, Amrita School of Engineering, Kollam, Kerala — CGPA: 8.48

Publication: R. Roy and G. Udupa, “Design and Analysis of Soft Prosthetic Hand based on Asymmetric Flexible Hydraulic Actuator,” IEEE IC3SIS, 2022. ieeexplore.ieee.org/document/9885620

Bachelor of Technology, Mechanical Engineering

2014 - 2018

Kerala University, Sree Buddha College of Engineering, Alappuzha, Kerala

Project: "Design and Fabrication of Fire Fighting Robot"

Higher Secondary - Science

2012 - 2014

Kerala State Board, SMHSS Patharam, Kollam, Kerala — Percentage: 96%

CERTIFICATIONS

- One Million Prompters — Dubai Centre for Artificial Intelligence (DCAI), Government of Dubai
- Google AI Essentials — Google
- Generative AI in Action — IBM
- Cloud Computing Fundamentals — IBM